



KCL AND KVL

CIRCUIT ANALYSIS METHOD

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TOPIC OUTLINE

Circuit Convention

Kirchhoff's Current Law (KCL)

Kirchhoff's Voltage Law (KVL)



CIRCUIT CONVENTION



CONVENTION

A convention is a widely accepted practice, method, or behavior that is followed by common agreement or tradition, rather than by formal rules.

Color coding in Offices

red – urgent documents

blue – general files

green – financial records

This is a common practice but not formally regulated.



STANDARD

A standard is a formal, established guideline, rule, or specification that is often mandatory and enforced by an authoritative body or organization.

IEC 60062 Resistor Color Code

black – 0

brown – 1

red – 2

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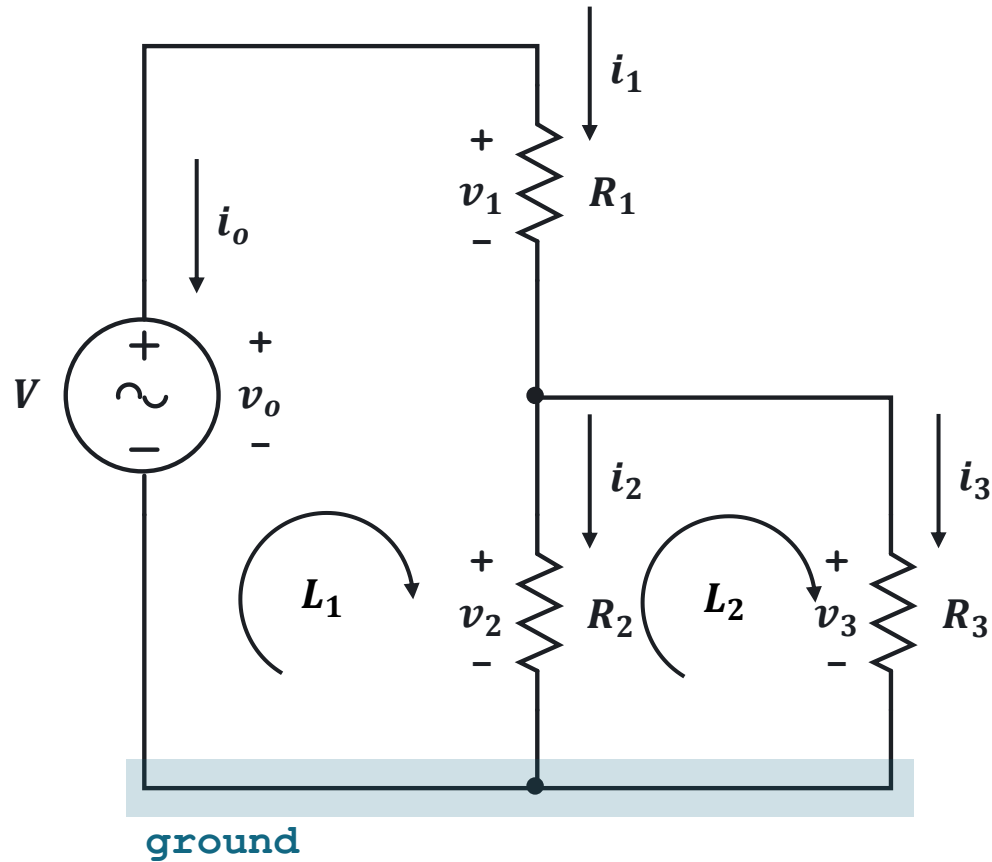
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white – 9

Resistors have colored bands that represent specific digits, multipliers, and tolerance values.



LABELING VARIABLES



Steps in Labeling Variables

1. Label the Reference Node (ground):

Select a reference node with the most connections or the negative (-) terminal of a voltage source.

2. Label Node Voltages:

Mark higher potentials as positive (+) relative to the reference node.

3. Label Currents:

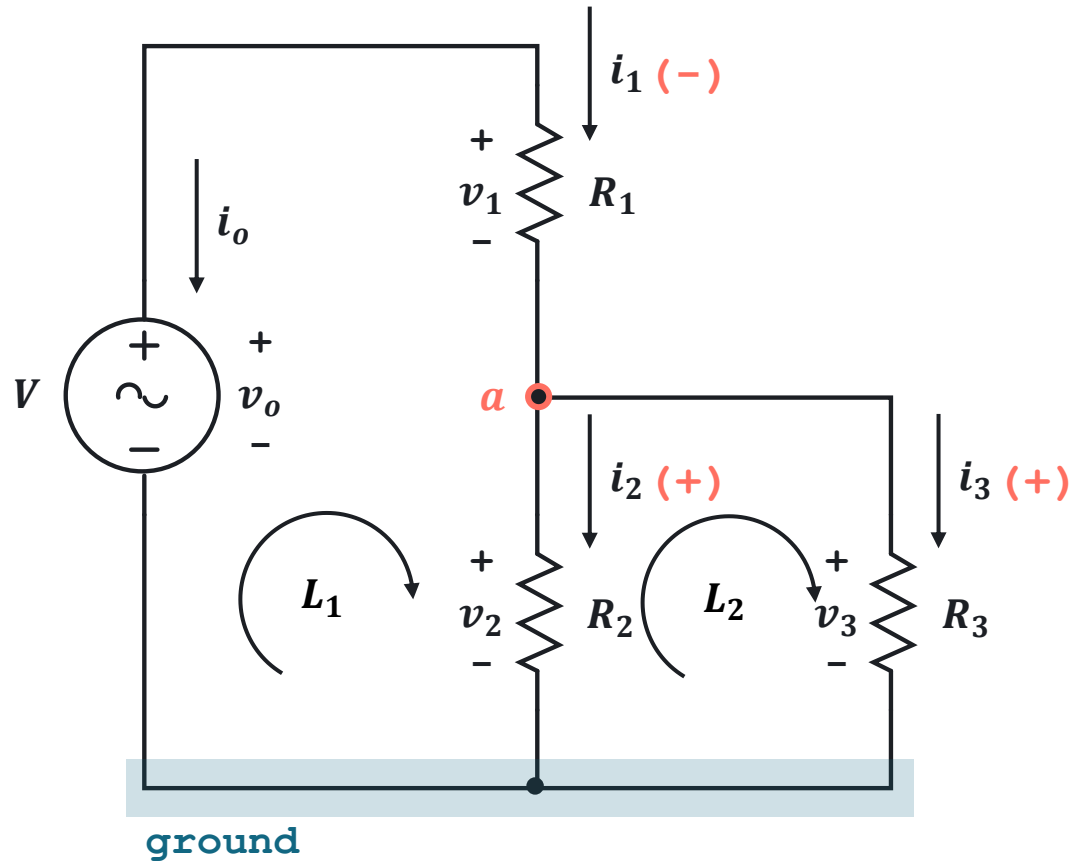
Entering the positive (+) terminal of a component.

4. Create a voltage loop:

Follow the defined current directions.



CURRENT FLOW CONVENTION



Current Flow Convention

- Current entering a node is negative $(-)$
- Current leaving a node is positive $(+)$

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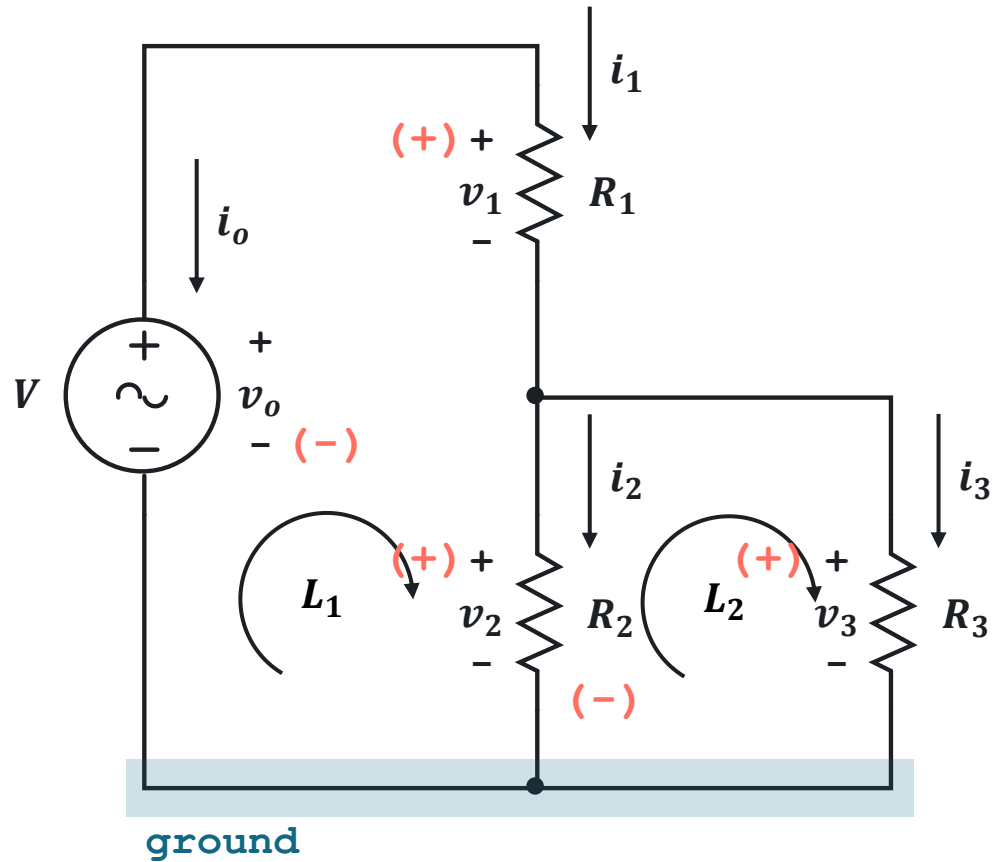
$-i_1$

$+i_2$

$+i_3$



VOLTAGE LOOP CONVENTION



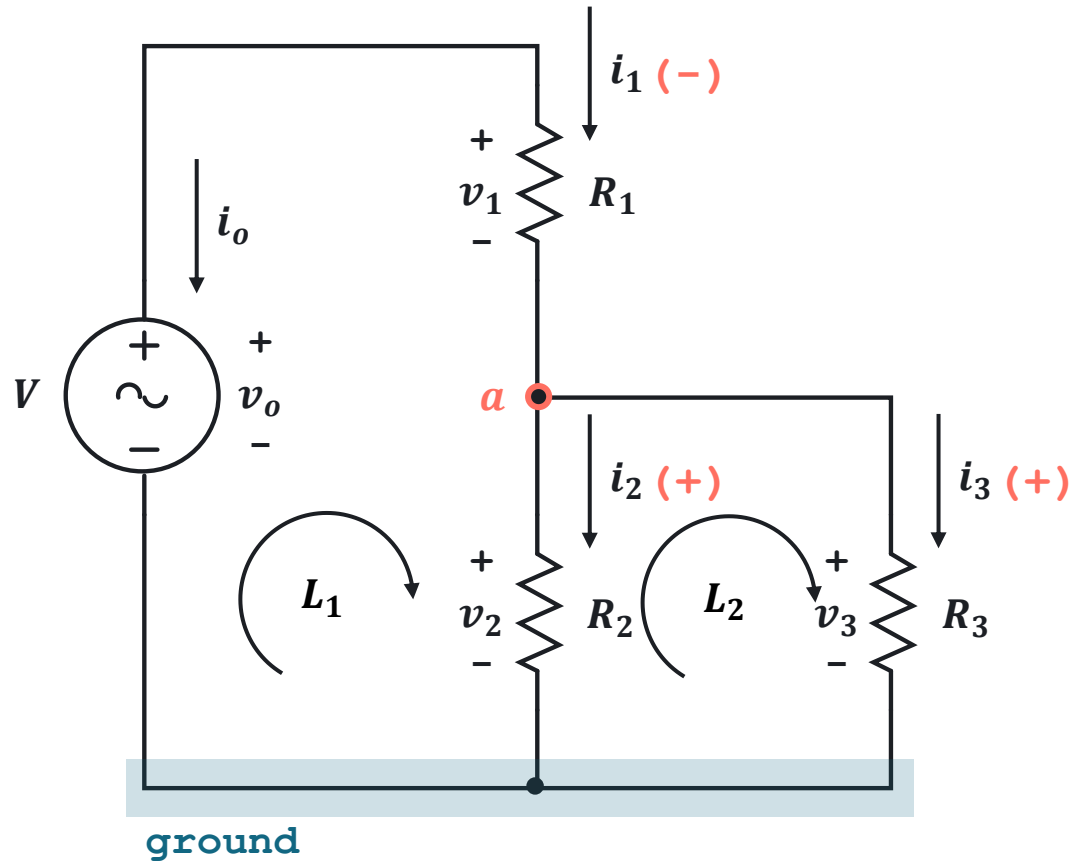
Voltage Loop Convention

The “sign” of voltage of the element is the first sign the loop encounters.

 $@L_1$
$$-v_o$$
 $+v_1$ $+v_2$ $@L_2$
$$-v_2$$
 $+v_3$ 

KIRCHHOFF'S CURRENT LAW AND VOLTAGE LAW





Kirchhoff's Current Law

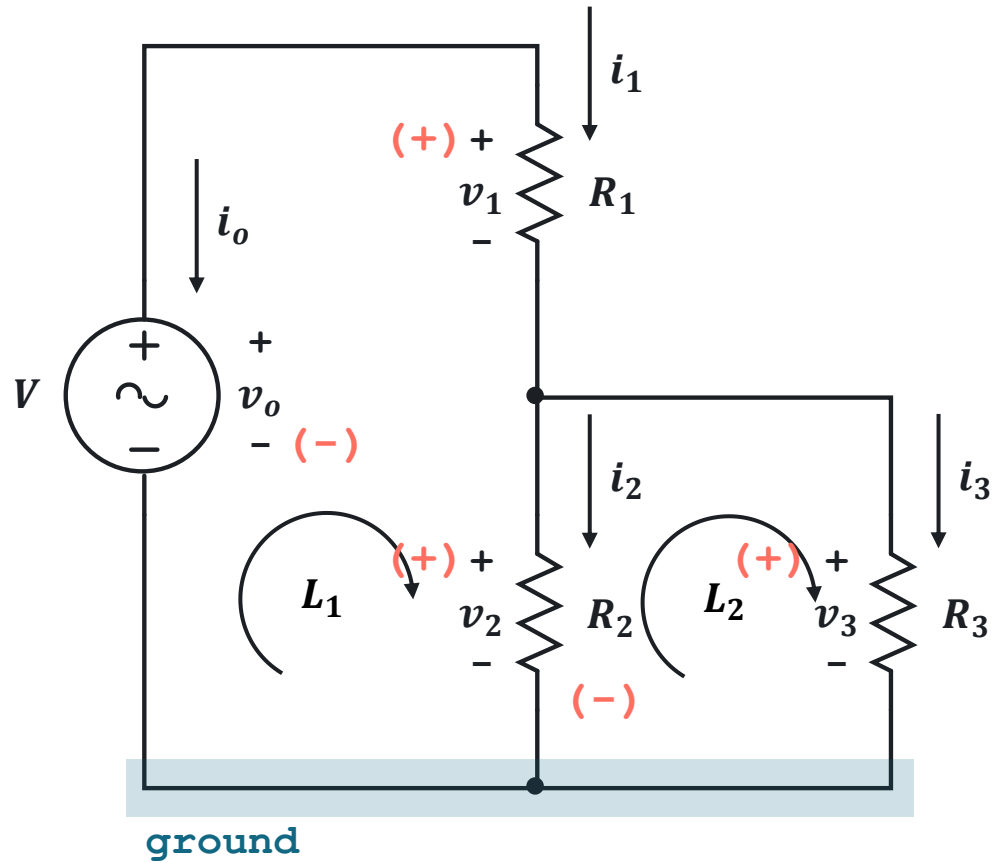
The summation of currents going-in and going-out a node is zero.

$$\sum i_j = 0$$

KCL @ a

$$-i_1 + i_2 + i_3 = 0$$





Kirchhoff's Voltage Law

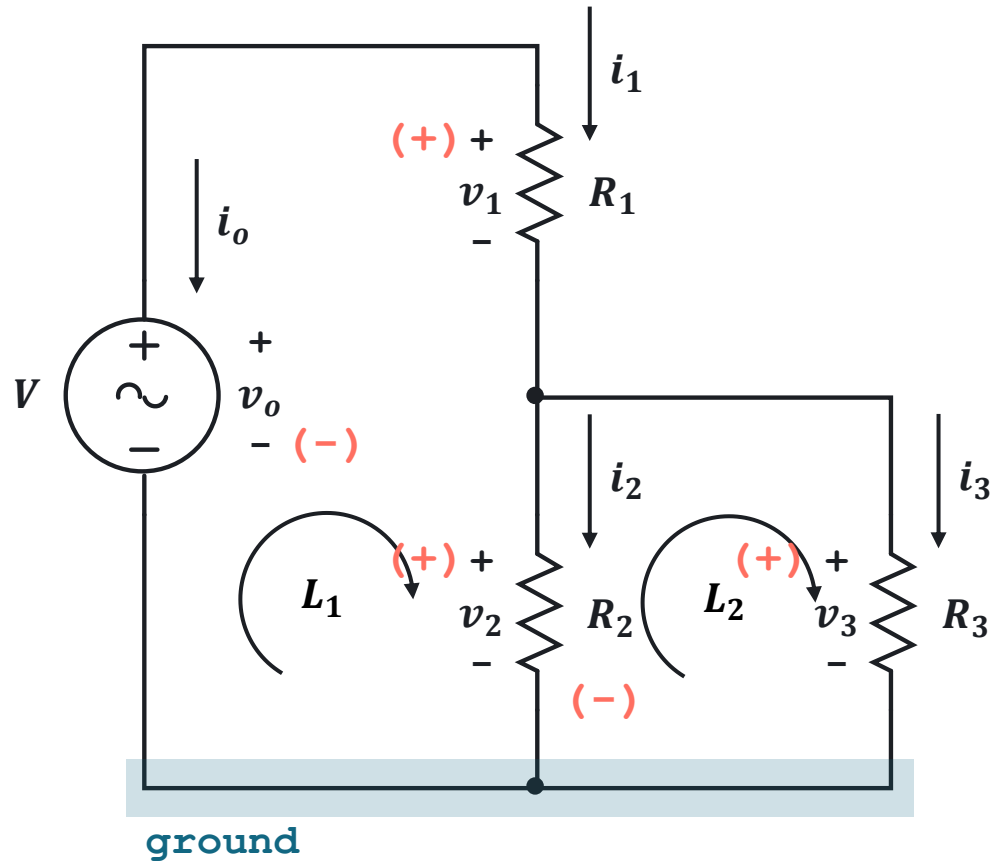
The summation of voltages in a closed-loop is zero.

$$\sum v_j = 0$$

KVL @ L_1

$$-v_0 + v_1 + v_2 = 0$$





Kirchhoff's Voltage Law

The summation of voltages in a closed-loop is zero.

$$\sum v_j = 0$$

KVL @ L_2

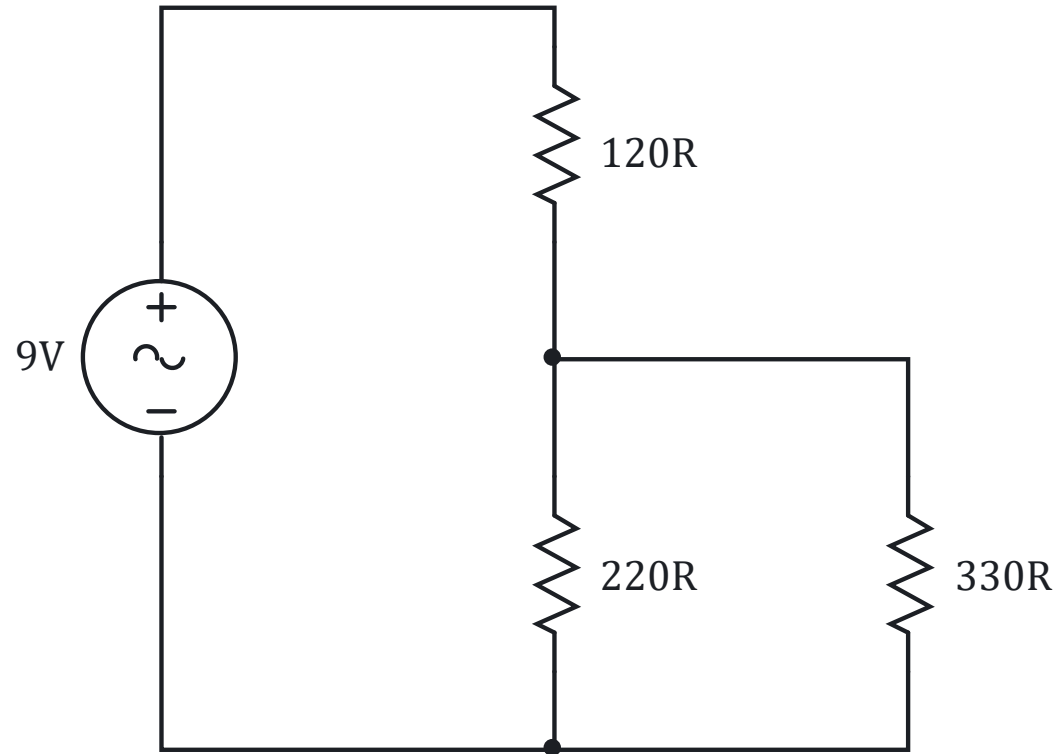
$$-v_2 + v_3 = 0$$



EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor in the given circuit.

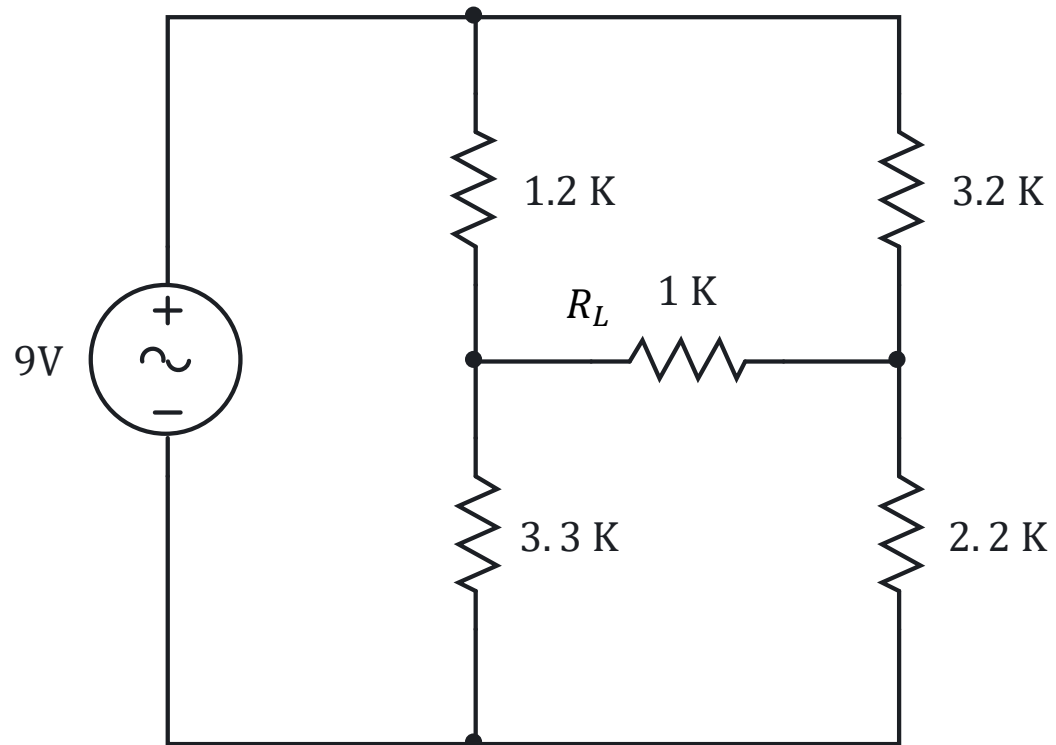
Solution



EXERCISE

Determine the load R_L voltage and current of the given circuit.

Solution



LABORATORY

